KEY POINTS

Short Textbook of Anesthesia

Source: Short Textbook of Anesthesia – Ajay Yadav (6th Edition)

38 Chapters • 598 Key Points

About this document:
- Contains all KEY POINTS from the end of each chapter.
- Organized chapter-wise for quick revision.
- Ideal for last-minute review before exams.

Chapter 1 – Applied Anatomy, Physiology and Physics

1.	Subglottis is the narrowest part in children up to the age of 6 years however, this does not act as deterrent to use cuffed tubes in children.
2.	During general anesthesia with muscle relaxants the cords are held in cadaveric position. SECTION 1: Fundamental Concepts 12
3.	Due to shorter, wider and less acute angle chances of endobronchial intubation are more on right side.
4.	Aspiration in supine position most commonly involves apical segment of right lower lobe.
5.	Maximum airway resistance to airflow occurs in trachea and then main bronchus and minimum in terminal bronchi.
6.	Endotracheal tubes and tracheostomy decrease the anatomical dead space by bypassing the upper airways.
7.	During general anesthesia alveolar dead space is increased.
8.	P50 i.e. partial pressure at which oxygen saturation is 50% is not affected by anesthetics.
9.	All inhalational agents blunt hypoxic pulmonary vasoconstriction (HPV) response thereby increasing the shunt fraction.
10.	Closing capacity is normally 1 L less than functional residual capacity.
11.	Venturi principle is often utilized in anesthesia particularly with Venturi masks.

Chapter 2 – Anesthesia Delivery Systems (Machine and Circuits)

1.	The color of oxygen cylinder is black body with white shoulders (top) and pressure is 2,000 pounds per square inch (psi).
2.	Nitrous oxide is stored in blue color cylinders at a pressure of 760 psi.
3.	Entonox is the mixture of 50% oxygen and 50% nitrous oxide.
4.	Pin index system is to prevent the wrong fitting of cylinders while diameter index safety system is to prevent the wrong fitting of central supply pipe lines.
5.	Fail safe valve, oxygen-nitrous oxide proportionater and oxygen tube placed most downstream are to decrease the possibility of delivery of hypoxic mixture to patient.
6.	Upper end of bobbin determines the flow rate in rotameter.
7.	Aladin cassette vaporizer is the latest vaporizer, which can be used to deliver all inhalational agents.
8.	Magill circuit is the circuit of choice for spontaneous ventilation. Fresh gas flow should be kept equal to minute volume to prevent rebreathing in a spontaneously breathing patient.
9.	Bain circuit is the circuit of choice for controlled ventilation. Fresh gas flow for controlled ventilation is 1.6 times of minute ventilation.
10.	Jackson Rees (Mapleson F) is the most commonly used semiclosed circuit for children.

11.	Sevoflurane by reacting with sodalime and barylime can produce compound A.
12.	Desiccated (dry) sodalime and Barylime (Barylime > Sodalime) can produces carbon monoxide with the following inhalational agents in decreasing order: Desflurane > > Isoflurane >> halothane = Sevoflurane.
13.	Desiccated sodalime can cause burns of respiratory mucosa with sevoflurane by producing hydrogen fluoride.
14.	Amsorb neither produces compound A nor carbon monoxide with desiccated sodalime and Barylime.

Chapter 3 – Airway Management and Intubation

1.	The most common airway used to prevent the tongue fall is Guedel's airway.
2.	The major limitation of mask ventilation is that it increases the chances of aspiration.
3.	Macintosh is the most commonly used laryngoscope.
4.	Intubation success rates of 94–99% has been reported for video laryngoscopy as a rescue modality after failed direct laryngoscopy.
5.	Most commonly used video laryngoscopes is C-MAC.
6.	Fiberoptic laryngoscope/bronchoscope-guided intubation is considered as gold standard technique for the management of difficult/failed intubation.
7.	Current day anesthetic practice is an era of supraglottic devices.
8.	Second generation LMA has shown significant decrease in incidence of aspiration as compared to first generation LMA.
9.	Among the second generation LMA, I-Gel is most frequently used in India.
10.	Definitive airway management devices includes endotracheal tube and tracheostomy.
11.	The chief advantage of PVC tube is that their cuff is low pressure and high volume.
12.	During laryngoscopy, there should be extension at atlanto-occipital joint and flexion at cervical spine.
13.	Capnography is the surest sign to confirm the correct position of endotracheal tube.
14.	The traditional approach of not using cuffed tubes in children up to 10 years does not hold true in present day practice.
15.	The most common cause of hypoxia for double lumen tube is ventilation perfusion mismatch.
16.	Fixed performance devices deliver accurate oxygen concentration.
17.	Heat and moisture exchanger (HME) is the most commonly used device to preserve humidity in anesthesia.

Chapter 4 – Preoperative Assessment and Premedication

1.	Preoperative assessment is done to obtain history, do physical examination, order investigations, risk stratification (by ASA grading) and give preoperative instructions.
2.	The decision to order preoperative tests should be guided by the patient's clinical history, comorbidities, physical examination findings and the proposed surgery. Only the investigations which are necessary should be done.
3.	For solid food 8 hours of fasting is must while clear fluids can be given up to 2 hours. Medications need to be stopped:
4.	MAO-A inhibitors (3 weeks before)
5.	Oral anticoagulants (Warfarin 4 days prior)
6.	Heparin (Low molecular weight 12 hours before)
7.	Antiplatelets except aspirin (Clopidogrel 5 days prior)
8.	Thrombolytic (10 days prior)
9.	Nonsteroidal anti-inflammatory drugs (NSAIDs) (48 hours prior if used with other antiplatets)
10.	High dose estrogen oral contraceptives (4 weeks prior)
11.	Viagra (24 hours prior)
12.	Disulfiram (10 days prior)
13.	All herbal medications (7 days before)
14.	Smoking (8 weeks before) Medications for which only morning dose to be omitted:
15.	ACE inhibitors and angiotensin II antagonist
16.	Oral hypoglycemics
17.	Topical creams and ointments
18.	Vitamins and iron. Dose adjustment is needed for:
19.	Cholinesterase inhibitors
20.	Corticosteroids
21.	Rest all medications are continued in the same dosages and same regime with morning dose on the day of surgery to be taken with a sip of water.
22.	In the current day practice of anesthesia patient is premedicated with a purpose not as a routine procedure.

Chapter 5 – Difficult Airway Management

1.	Airway assessment is one of the most important parts of preoperative assessment, failure to do so can make anesthetist negligent in case of some catastrophe.
2.	Mallampati grading, thyromental distance and neck movements are the three most important parameters to be assessed in every case.
3.	Video laryngoscopy have revolutionized the practice of airway management; intubation success rates can be as high as 94–99% with video laryngoscopy after failed intubation with direct laryngoscopy.
4.	Every head injury should be considered to be having cervical injury until proved otherwise.
5.	For cervical spine injury intubation should be done after neck stabilization with neck in neutral position.

Chapter 6 – Monitoring in Anesthesia

1.	In spite of availability of sophisticated monitors nothing can replace the vigilance of an anesthetist.
2.	The most preferred leads for ECG monitoring are lead II and V5. Arrhythmia are best detected in lead II and ischemia's in lead V5.
3.	The maximum interval between two blood pressure recordings should not exceed more than 5 minutes.
4.	Invasive blood pressure monitoring is considered as gold standard monitoring of blood pressure.
5.	As radial artery has collaterals (transpalmar arch) for the blood supply of hand, it is most commonly chosen artery for cannulation.
6.	Measuring dynamic parameters like stroke volume variation by arterial cannulation may be considered as one of the most recent and useful advancement in the field of monitoring.
7.	PAOP is best utilized to differentiate between cardiogenic and noncardiogenic pulmonary edema (ARDS).
8.	Oxygen saturation of mixed venous blood is best guide to assess tissue perfusion (cardiac output).
9.	In present day medicine transesophageal echocardiography (TEE) and transthoracic echocardiography (TTE) are the best tools to assess cardiac function and detect wall motion abnormality, i.e. Ischemia in perioperative period.
10.	Transesophageal echocardiography provides the best window for measuring and assessing cardiac functions in intraoperative period.
11.	Co-oximeters are the special type of oximeters which can differentiate between normal and abnormal hemoglobin.
12.	In current day practice all monitors use infrared absorption technique to measure concentration of all expired gases except oxygen and nitrous oxide.
13.	Capnography is the surest technique to confirm intubation.
14.	Electrical impendence pulmonometery is the simplest and most commonly used method to detect apnea in a non-intubated patient.
15.	Hypothermia is the most common thermal perturbation seen during anesthesia.

16.	Temperature should be monitored in patients receiving general anesthesia >30 minutes and in all patients whose surgery lasts > 1 hour (irrespective of type of anesthesia).
17.	Lower end of esophagus is considered as best site for core body temperature measurement. However, the most accurate measurement of core body temperature is provided by pulmonary artery.
18.	Each degree (°C) fall in temperature reduces the metabolic rate by 6 to 7%.
19.	Most commonly used muscle for neuromuscular monitoring is adductor pollicis supplied by ulnar nerve.
20.	Train of four is the most commonly used modality for neuromuscular monitoring in clinical practice.
21.	Fading, post-tetanic facilitation and post-tetanic count are only exhibited by non-depolarizing muscle relaxants.
22.	Bispectral index (BIS) monitor is the first scientifically validated and commercially available monitor to assess the depth of anesthesia.

Chapter 7 – Fluids and Blood Transfusion

1.	Crystalloids should be replaced in a ratio of 1.5:1 to intravascular volume loss while colloids in a ratio of 1:1.
2.	Colloids in high doses can interfere with clotting and causes renal dysfunction.
3.	Ringer lactate is the most preferred fluid for maintenance as well as replacement.
4.	Children are prone for fluid overload therefore should receive 2/3rd of the calculated dose.
5.	Colloids can be safely given even if there is endothelial injury like in ARDS or head injury.
6.	The main aim of blood transfusion is to improve the oxygen delivery of tissues, not the volume replacement (which should be done by fluids).
7.	In spite of varying controversies the most acceptable trigger point for blood transfusion are blood volume loss >20% and hemoglobin < 8 gm%.
8.	One unit of blood/packed cells increases hemoglobin by 0.8%.
9.	Most common cause of mismatch transfusion is ABO incompatibility.
10.	DIC is the most common cause of death after massive transfusion while acute lung injury remains the leading cause of mortality after transfusion.
11.	Massive transfusion related dilutional coagulopathies should not be treated prophylactically.
12.	Platelets are the only blood products which are stored at room temperature.
13.	Normovolemic/isovolemichemodilution is the most widely used technique for autologous blood donation.

Chapter 10 – Inhalational Agents

1.	Inhalational agents are mainly used in anesthesia for maintenance.
2.	The major changed happened in mechanism of action of inhalational agents is shifting of mechanism from lipid sites to protein sites.
3.	Best estimate for determining the potency of inhalational agent is minimum alveolar concentration (MAC). Halothane is the most potent while nitrous oxide is the least potent among the agents used nowadays.
4.	Blood gas partition coefficient indicates induction and recovery. Among the agents used nowadays induction is fastest with desflurane and slowest with Halothane.
5.	All inhalational in current use decreases cardiac output and depresses respiration.
6.	Halothane (and theoretically other agents too except Isoflurane) can sensitize myocardium to adrenaline, exogenous to heart.
7.	Direct hepatocellular damage is seen with halothane (and theoretically with Isoflurane and Desflurane) but not with sevoflurane.
8.	Inflammable agents used in past were Ether and Cyclopropane.
9.	Sevoflurane can produce Compound A with sodalime and Barylime.
10.	Desflurane (and other agents too) can produce Carbonmonoxide by reacting with desiccated sodalime and Barylime.
11.	Sevoflurane by reacting with desiccated sodalime and barylime can cause burns of respiratory mucosa.
12.	times more solubility of nitrous oxide than nitrogen makes it absolutely contraindicated in pneumothorax, pneumopericardium, pneumoperitoneum and pneumoencephalus.
13.	Nitrous oxide can deplete ozone layer and produce global warming.
14.	Immunologic mechanism of halothane hepatitis warrants to avoid frequent use of halothane; at least an interval of 3 months is mandatory between two exposures.
15.	Isoflurane is the agent of choice for cardiac patients.
16.	In the current day practice Sevoflurane is preferred over Isoflurane for neurosurgery.
17.	Desflurane is agent of choice for prolonged surgeries, old age, renal diseases, obesity, day care surgery and shock.
18.	Sevoflurane is the agent of choice for induction in children.
19.	Sevoflurane is the agent of choice for hepatic patients.
20.	At higher concentrations, sevoflurane can produce convulsions.
21.	Enflurane was highly epileptogenic.
22.	Ether is safest, cheapest and only complete anesthetic while on flip side, it is highly inflammable and explosive and induction and recovery are very unpleasant.

23.	Methoxyflurane can produce vasopressin resistant polyuric renal failure.
24.	Trielene by reacting with sodalime can produce dichloroacetylene (craniotoxic) and phosgene (ARDS).

Chapter 12 – Intravenous Anesthetics

1.	Most commonly used IV anesthetic for induction in current day practice is propofol.
2.	Although elimination half-life of Thiopentone is 10.4 hours but consciousness is regained after 15–20 minutes due to redistribution.
3.	Thiopentone can be utilized for cerebral protection in case of focal ischemia.
4.	Thiopentone high alkalinity is responsible for local complications like intra-arterial injection.
5.	Vasospasm and thrombus formation are the major events responsible for causing gangrene after intra-arterial injection of Thiopentone.
6.	Thiopentone is absolutely contraindicated in acute intermittent and variegate porphyria but can be used safely in porphyria cutaneatarda.
7.	Injection of propofol is painful.
8.	As propofol contains egg lecithin there are high chances of contamination of Propofol. CHAPTER 12: Intravenous Anesthetics 121
9.	Propofol is the IV agent of choice for day care surgery.
10.	Propofol is a potent antiemetic agent.
11.	Etomidate is the most cardiovascular stable among all IV agents.
12.	The major side effect of etomidate is adrenocortical suppression.
13.	Midazolam is the only BZ used in intraoperative period.
14.	Ketamine is the most potent analgesic among IV agents.
15.	Emergence reactions, mainly hallucinations, are the major limiting factor for restricting the use of ketamine.
16.	All pressures like intraocular, intragastric, intracranial are raised by ketamine.
17.	Ketamine is the induction agent of choice for shock, full stomach, low cardiac output states and right to left shunts.
18.	The most important receptors for the action of opioids are mu receptors.
19.	The most important characteristic of agonist- antagonists is ceiling to respiratory depression.

20.	Recurring respiratory depression is exhibited by all pure agonists.
21.	Muscle rigidity (Wooden chest syndrome) is most commonly seen with alfentanil.
22.	Tolerance is seen to all actions of opioids except constipation and miosis.
23.	The chronic use of opioids can produce hyperalgesia.
24.	Sufentanil is the agent of choice for inhibiting stress response to laryngoscopy and intubation.
25.	Remifentanil is the opioid of choice for day care surgery.
26.	Methylnaltrexone and Alvimopan are the peripheral opioid antagonist which can reverse the peripheral side effects without reversing the analgesia.
27.	Dexmedetomidine is a new $\alpha 2$ agonist which is frequently used in anesthetic practice.

Chapter 13 – Muscle Relaxants

1.	In clinical practice central muscle are blocked and recover earlier than peripheral muscles (limb muscles).
2.	Because of early onset and short duration Succinylcholine is the ideal muscle relaxant for intubation.
3.	Hyperkalemia is the most prominent effect of suxamethonium.
4.	Succinylcholine is the most commonly implicated drug for malignant hyperthermia.
5.	Intraocular, intragastric and intracranial pressures are increased by suxamethonium.
6.	Fading after suxamethonium is pathognomonic of phase II block.
7.	Continue intermittent positive pressure ventilation and wait for spontaneous recovery is the treatment of choice for low and atypical pseudocholinesterase.
8.	Benzylisoquinoline releases histamine while steroidal compounds are vagolytic.
9.	Onset of a nondepolarizer is inversely proportional to potency.
10.	Vecuronium is most cardiovascular stable muscle relaxant.
11.	Rocuronium is nondepolarizer of choice for intubation.
12.	Intense bronchospasm led to withdrawal of rapacuronium from market.
13.	Atracurium and cisatracurium are metabolized by Hoffman degradation making them relaxant of choice for hepatic and renal failure.
14.	Cisatracurium does not release histamine and Laudonosine production is 5 times less than atracurium.

15.	Mivacurium is the muscle relaxant of choice for day care surgery.
16.	Gallamine is the only nondepolarizer which can cross the placenta therefore absolutely contraindicated in pregnancy.
17.	Gantacurium onset and duration is almost comparable to suxamethonium.
18.	To prevent muscarinic side effects, anticholinergic like atropine or glycopyrrolate has to be given with cholinesterase inhibitors.
19.	Sugammadex is the newer reversal agent which rapidly and completely reverses the block by steroidal type of nondepolarizers but cannot reverse benzylisoquinoline compounds.
20.	Sustained jaw clench over a tongue blade is considered as best clinical sign however train of four ratio > 0.9 guarantees recovery.

Chapter 14 – Perioperative Complications of GA

1.	Maximum complications are reported in maintenance period however more than 80% of cardiac arrest occurs at the time of induction.
2.	Postoperative respiratory depression has been cited as the most common cause of anesthesia related death in perioperative period.
3.	Death is the leading cause of claims as per American Society of Anesthesiologists (ASA) claim studies.
4.	Aspiration is considered as preventable complication of anesthesia.
5.	Full stomach is the single most important predisposing factor for aspiration.
6.	Rapid sequence induction technique must be employed for giving GA to the patients who are at high risk of aspiration.
7.	Atelectasis is the most common postoperative pulmonary complication and the most common cause of atelectasis is secretions.
8.	Use of oxygen analyzers is mandatory as per ASA task force guidelines.
9.	Secretions is the most common cause of laryngospasm in anesthesia.
10.	Postoperative hypertension and tachycardia are more responsible for unplanned ICU admission and carries higher morbidity and mortality than hypotension and bradycardia.
11.	Tachycardia is the most common arrhythmia seen in perioperative period.
12.	The most common cause of hypotension in perioperative period is intravascular volume depletion.
13.	Nitroglycerin as continuous infusion is the most commonly used agent to produce controlled hypotension.
14.	Anesthetic drugs which can cause convulsion are—Local anesthetics, methohexitone, enflurane, sevoflurane and atracurium/cisatracurium.
15.	Delayed recovery is the most common CNS complication.

16.	Brachial plexus injuries are the most common postoperative nerve injury associated with general anesthesia followed by ulnar nerve. However after regional anesthesia lumbosacral nerve roots are most commonly involved nerves.
17.	Ondansetron/Granisetron remains the drugs of choice for treatment as well as prophylaxis for postoperative nausea and vomiting.
18.	Pain is the most common complication is postoperative period.
19.	Hypothermia is the most common thermal perturbation seen in anesthesia.
20.	Succinylcholine is the most commonly implicated drug while halothane is the most commonly implicated inhalational agent for causing malignant hyperthermia.
21.	Hyperkalemia induced ventricular fibrillation is the most common cause of death in malignant hyperthermia.
22.	Propofol and non-depolarizing muscle relaxants can delay or prevent malignant hyperthermia.
23.	Muscle relaxants are the most common cause of anaphylaxis followed by latex allergy.
24.	Cautery is responsible for 68% of fires seen in OT.

Chapter 15 – Local Anesthetics

1.	Local anesthetics are classified on the basis of chemical structure and duration of action.
2.	Chloroprocaine is the shortest acting and dibucaine is the longest acting local anesthetic.
3.	Recent studies have proven that local anesthetics not only act on sodium channels, but also block potassium and calcium channels.
4.	Sequence of blockade of nerve fibers is A > B> C. The traditional notion that thin fibers are most sensitive to the action of local anesthetics does not hold true in present day practice.
5.	Potency is proportional to the duration of action.
6.	Maximum systemic absorption has been seen after intercostal nerve block.
7.	Central nervous system is the first system involved in local anesthetic toxicity.
8.	At clinically used doses all local anesthetics are vasodilators except cocaine, levobupivacaine and ropivacaine.
9.	Allergic reactions are common with esters but rare with amides.
10.	Methemoglobinemia is usually seen with prilocaine however benzocaine can also cause methemoglobinemia.
11.	Maximum local neurotoxicity has been reported with chloroprocaine.
12.	Local anesthetics with adrenaline can cause necrosis and gangrene if used for ring block of fingers, toes, penis or pinna because these structures have end arteries.

13.	For spinal and epidural anesthesia preservative free preparations should be used.
14.	Maximum safe dose of lignocaine without adrenaline is 4.5 mg/kg (maximum 300 mg) while with adrenaline it is 7 mg/kg (maximum 500 mg).
15.	Wide differential blockade of bupivacaine and ropivacaine makes them local anesthetic of choice for postoperative pain relief and painless labor.
16.	Prilocaine has significant extrahepatic metabolism.
17.	Because of high cardiotoxicity bupivacaine is absolutely contraindicated for bier's block.
18.	Amiodarone now a days is considered as drug of choice for the treatment of bupivacaine (and other local anesthetic) induced ventricular tachycardia.
19.	Cardiotoxicity and CNS toxicity of ropivacaine is far lesser than bupivacaine and lesser than levobupivacaine (but still higher than lignocaine).
20.	Mepivacaine, bupivacaine, levobupivacaine and ropivacaine cannot be used while procaine and chloroprocaine should not be used for surface anesthesia. Nerve blocks may be classified as central neuraxial blocks (spinal and epidural) and peripheral nerve blocks. Nerve blocks are given either for regional anesthesia or for acute and chronic pain relief. (For pain relief blocks see Chapter 39, page no. 273). TECHNIQUE The conventional method of giving a nerve block is by eliciting paresthesia and/or using the nerve stimulator. However, the use of ultrasound has revolutionized the practice of nerve block in modern day anesthetic practice. BLOCKS OF UPPER LIMB Brachial Plexus Block This is the second most commonly performed block (after central neuraxial block). It is used for surgery of upper limb and shoulder. Brachial plexus can be blocked by 4 approaches— interscalene, supraclavicular, infraclavicular and axillary approach. Interscalene Approach The usual indication for interscalene block is surgery on the shoulder or upper arm. In this technique brachial plexus is blocked between anterior and middle scalene at the level of cricoid. Blockade occurs at the level of the superior and middle trunks. Ulnar nerve is usually spared with this approach making it unsuitable for hand surgeries. Complications: z z Phrenic nerve block: Phrenic nerve block occurs in almost all patients but unilateral phrenic nerve block has no clinically significant consequences in a normal individual (pulmonary functions are only reduced by 20–25%). z z Horner syndrome (due to block of stellate ganglion). z z Epidural and intrathecal injection can be dreadful complication. z z Pneumothorax if the needle insertion is too low. z z General complications like nerve injury, neuritis, intravascular injection, bleeding and hematoma formation, infection or injury to a nearby structure and systemic toxicity of local anesthetic. Supraclavicular Approach This is very commonly used approach. This approach is usually employed for surgeries on lower arm, elbow, forearm and hand. Distal trunk and proximal division of brachial plexus is blocked by this approach. Technique: Patient lies supine with a small wedge under the shoulder. The arm is extended and adducted; needle is inserted at a point 1 cm superior to midpoint of clavicle after palpating the subclavian artery. The needle is inserted lateral to subclavian vessels (Fig. 16.1). The needle is inserted in downward and backward direction till paresthesia is elicited. After eliciting paresthesia 20–30 mL of 1–1.5% xylocaine alone or mixed with bupivacaine is injected. Complications: z z Pneumothorax: Dome of pleura can be punctured. Incidence is 1–6% but fortunately in more than 98% of the cases pneumothorax C H A P T E R 16 Peripheral Nerve Blocks

Chapter 16 – Peripheral Nerve Blocks

1.	The use of ultrasound has not only made the nerve blocks technically easy but also has improved the precision.

2.	Ulnar nerve is usually spared with interscalene approach while musculocutaneous is usually spared with axillary approach.
3.	The most bothersome complication of supraclavicular block is pneumothorax. Keeping the direction of needle lateral to the Subclavian artery is the best way to avoid pneumothorax.
4.	Only safe local anesthetics, i.e. lignocaine and prilocaine are permitted for Bier block. Drug with high toxicity like bupivacaine is absolutely contraindicated.
5.	Cervical plexus block is best utilized for carotid endarterectomy.
6.	Maximum systemic absorption of local anesthetic occurs after intercostal nerve block.
7.	Associated coagulopathy and infection at needle site are the absolute contraindication for nerve blocks.

Chapter 17 – Central Neuraxial Blocks

1.	The line joining the highest point on iliac crest corresponds to L4 and L5 interspace or L4 spine.
2.	Spinal cord extends from medulla oblongata to lower border of L1 in adults and L3 in infants.
3.	The sequence of blockade after CNB is autonomic > sensory > motor.
4.	Several meta-analyses have shown a reduced risk as high as 30% in overall mortality and morbidity in patients who underwent surgeries under CNB as compared to GA.
5.	Apnea after spinal anesthesia is usually due to severe hypotension.
6.	The most common cause of nausea and vomiting is hypotension.
7.	Hypotension is the most common complication of spinal anesthesia.
8.	Urinary retention is the most common postoperative complication of central neuraxial blocks in present day practice.
9.	Hyperbaric bupivacaine is the most commonly used drug for spinal.
10.	Dose and Baricity are 2 very important determinants of level of block.
11.	Ephedrine is the most preferred vasopressor for the treatment of spinal hypotension.
12.	High spinal is associated with bradycardia and hypotension.
13.	The drug of choice for treatment of shivering is pethidine.
14.	Postspinal headache is low pressure, meningovascular headache.
15.	Postural relation, i.e. pain in sitting and standing position and getting relived in lying down is diagnostic of postspinal headache. SECTION 6: Regional Anesthesia 180
16.	Use of small gauge and dura separating needles are the most important preventive measures to prevent post spinal headache.
17.	6th is most commonly involved cranial nerve after spinal anesthesia.

18.	Streptococcus viridans has been cited as the most common organism responsible for meningitis after spinal while Staphylococcus epidermidis is the most common organism responsible for meningitis after epidural.
19.	Transient neurological symptoms and cauda equina syndrome lead to almost elimination of lignocaine for spinal.
20.	The most commonly used needle for epidural is Tuohy needle and most commonly used method to locate epidural space is by loss of resistance technique.
21.	Fentanyl + bupivacaine or ropivacaine as continuous infusion is the most commonly used combination for postoperative analgesia and painless labor.
22.	There is less hypotension in epidural as compared to spinal.
23.	Chances of infectious complications, epidural hematoma and neurotoxicity is more with epidural as compared to spinal.
24.	It is considered safe to give central neuraxial block 12 hours after last dose of LMWH and LMWH can be given safely given 6 hours after central neuraxial blocks.
25.	Patients on aspirin and NSAIDs can be safely given central neuraxial blocks while CNB should be delayed for 7 days after clopidogrel.
26.	Patients with platelet count more than 80,000/cubic mm can be safely given central neuraxial blocks.

Chapter 18 – Anesthesia for Cardiovascular Diseases

1.	Effort tolerance > 4 METS is generally associated with lesser complications.
2.	Beta blockers should not be started afresh in a patient who is not on beta blockers unless they can be started at least 1 week before surgery. CHAPTER 18: Anesthesia for Cardiovascular Diseases (For Noncardiac Surgeries) 191
3.	Aspirin should be continued while Clopidogrel should be stopped 5 days before surgery.
4.	If a patient had myocardial infarction/ischemia then elective surgery must be deferred for 6 weeks if the patient was treated without stenting, for 6 weeks if bare metal stent was used, for 6 months if new generation drug eluting stent was used, for 1 year if conventional drug eluting stent was used and for 6 weeks if treated with coronary artery pass graft (CABG).
5.	If a patient with drug eluted stent has to undergo emergency surgery then continuing antiplatelet therapy is always preferred.
6.	Transesophageal echocardiography(TEE) is the earliest and most sensitive monitor to detect intra-operative infarction.
7.	The most important goal of anesthetic management for patients of ischemic heart disease is to avoid a imbalance between myocardial oxygen demand and supply.
8.	Etomidate is the induction agent of choice for cardiac patients.
9.	Cardiac patients with normal left ventricular function can safely receive inhalational agents however if left ventricular function is compromised then anaesthesia should preferably be maintained with opioids.
10.	Vecuronium is the muscle relaxant of choice for cardiac patients.

11.	Central neuraxial blocks (CNB) should be avoided for fixed cardiac output lesions like mitral or aortic stenosis.
12.	CPR may be ineffective in patients with severe aortic stenosis.
13.	Contrary to the previous recommendation of antibiotic prophylaxis to all patients of prosthetic heart valves undergoing any procedure the new guidelines recommends antibiotics prophylaxis only for dental procedures which involves gingiva, oral mucosa and periapical region of teeth and invasive procedure that involve incision on infective tissue.
14.	The basic aim of anesthetic management of patients with Left to Right Shunts is to maintain low vascular resistance (hypotension) and maintain high pulmonary vascular resistance while for Right to Left Shunts is to maintain high vascular resistance (hypertension) or decrease pulmonary vascular resistance.
15.	As ketamine increases the systemic vascular resistance it is agent of choice for induction in right to left shunts.
16.	Elective surgery is contraindicated in patient with heart failure.
17.	Cardiac implanted electronic devices (CIED) should be programmed to asynchronous mode before surgery and reset to their original settings as soon as the surgery is over.
18.	Permanently implanted cardioverter-defibrillator (ICD) must be turned off before surgery.
19.	Although there is no universal standard cut off limit of BP to delay surgery however a general consensus is to delay elective surgery if diastolic blood pressure is > 110 mm Hg.
20.	All Antihypertensives should be continued except ACE inhibitors and angiotensin II receptor antagonist which are stopped on the day of surgery.
21.	Patient in shock must be resuscitated to at least mean arterial pressure > 65 mm Hg or systolic pressure > 90 mm Hg before considering for anesthesia.
22.	Central neuraxial blocks are absolutely contraindicated for severely hypovolemic and relatively contraindicated for mild to moderate hypovolemic patient.
23.	Ketamine is the induction agent of choice for shock.

Chapter 19 – Anesthesia for Respiratory Diseases

1.	Dyspnea at minimal activity and site of surgery are the two most important predictors of postoperative complications.
2.	Always rule out associated cor pulmonale in patients with pulmonary disease.
3.	Ideally smoking should be stopped 8 weeks before surgery however stopping smoking 12 hours before may be beneficial by reducing carboxyhemoglobin levels.
4.	The routine use of pulmonary function tests is not recommended.
5.	Regional anesthesia is always preferred over general anesthesia for respiratory compromised patients.
6.	Ventilation principle for COPD/asthma patients includes low tidal volume (6–8 mL/kg), low I: E ratio and slow respiratory rate of (6–10 breaths/min.) to allow adequate exhalation and prevent air trapping.

7.	Thoracic epidural is the most preferred approach to achieve analgesia for thoracic and upper abdominal surgeries while parenteral opioids should be avoided as much as possible.
8.	Ketamine in spite of its best bronchodilator property is generally avoided for asthma patients; its use is restricted to patients in active asthma (wheezing) undergoing life-threatening emergency surgeries.
9.	Propofol is preferred over ketamine for asthma patients in present day anesthesia practice.
10.	Thiopentone can induce bronchoconstriction therefore, should be avoided.
11.	Sevoflurane is the inhalational agent of choice for asthma patients.
12.	Intraoperative bronchospasm should first be treated by increasing the depth of anesthesia with inhalational agent (preferably sevoflurane).
13.	Nitrous oxide should not be used in emphysema.
14.	Elective surgery should be deferred in a patient of active (open) pulmonary tuberculosis.
15.	Elective surgery is contraindicated in patients with lower respiratory tract infection (lung infections). The decision to go ahead with elective surgery in patient with upper respiratory tract infection (URTI) depends on the severity of the disease.
16.	If surgery has to be deferred in a patient of URTI then it should ideally be deferred for 6 weeks.
17.	For patient of URTI general anesthesia (GA) with laryngeal mask airway (LMA) is always preferred over GA with intubation.

Chapter 20 – Anesthesia for CNS Diseases

1.	Levodopa should not be stopped before surgery.
2.	Dopamine antagonists like droperidol and metoclopramide should not be used in patients with Parkinson’s disease.
3.	The rising concern that general anesthesia worsens the dementia is not yet fully substantiated by studies.
4.	All antiepileptic should be continued.
5.	Thiopentone is the induction agent of choice for epileptics.
6.	After cerebrovascular accident elective surgery should be deferred for 9 months.
7.	Spinal anesthesia (but not epidural) and stress can precipitate multiple sclerosis; peripheral blocks can be given safely.
8.	Regional anesthesia and deep GA can prevent autonomic hyper-reflexia.
9.	All antidepressants including monoamine oxidase (MAO) inhibitors should be continued.
10.	Methohexital is the induction agent of choice for electroconvulsive therapy.

11.	Current recommendation is to continue lithium.
12.	Patients must not be taken for elective surgeries in acute drug withdrawal.
13.	Acute drug intake decreases while chronic intake increases the anesthetic requirement.

Chapter 21 – Anesthesia for Hepatic Diseases

1.	Child-Pugh classification is the gold standard scoring system to assess the severity of hepatic disease and hence the risk assessment.
2.	Elective surgery should be deferred in acute liver disease till the liver functions are stabilized.
3.	Due to the possibility of existing coagulopathy and intravascular volume depletion, central neuraxial blocks should be avoided in hepatic patients.
4.	Propofol, due to its short half-life and extrahepatic metabolism is the induction agent of choice for hepatic patients.
5.	Sevoflurane is the inhalational agent of choice for hepatic patients while halothane and nitrous oxide should be avoided.
6.	Atracurium and Cis-atracurium are the muscle relaxants of choice for hepatic patients.
7.	Albumin is the preferred colloid for a patient with compromised hepatic functions.
8.	As vecuronium has 30–40% excretion through bile, it should be avoided for patients with biliary obstruction.

Chapter 22 – Anesthesia for Renal Diseases and Electrolyte Imbalances

1.	Patients with acute kidney injury (AKI) should not undergo elective surgery until the injury is fully resolved.
2.	Dialysis should be performed within 24 hours before surgery.
3.	Correction of hyperkalemia is of utmost importance in chronic kidney disease patients.
4.	Maintaining fluid balance is essential in renal patients; excess fluid can precipitate pulmonary edema and fluid deficiency can worsen kidney injury.
5.	If invasive blood pressure monitoring is required then radial and ulnar artery should be spared as they may be required for hemodialysis in future.
6.	Central neuraxial blocks should be avoided due to associated uremic coagulopathy, uremic polyneuropathy and intolerance of renal patients to extra fluids to treat hypotension.
7.	Desflurane is the inhalational agent of choice because it does not produce fluoride.
8.	Sevoflurane should not be used in renal patients as it produces high levels of fluoride approaching renal threshold of 50 µmol/L.

9.	Non-depolarizing relaxant of choice is atracurium/Cis-atracurium while suxamethonium is contra-indicated due to associated hyperkalemia.
10.	Ringer lactate is preferred over normal saline with close monitoring of potassium levels. Colloids (hydroxyethyl starch) have been found to exacerbate renal injury.
11.	TURP syndrome is exhibited as fluid overloading, dilutional hyponatremia and hypo-osmolality.
12.	Hyperkalemia must be corrected before taking the patient for surgery while there is no need to correct mild chronic hypokalemia.
13.	Hypermagnesemia potentiates the action of both depolarizing and non-depolarizing muscle relaxants.

Chapter 23 – Anesthesia for Endocrinal Disorders

1.	Ruling out autonomic neuropathy is one of the most important part of preoperative assessment in diabetic patient.
2.	Regular insulin by continuous infusion is the most recommended method to control intraoperative plasma glucose.
3.	The target plasma sugar level should be between 150–180 mg/dL in perioperative period.
4.	Myocardial ischemia is the most common cause of death in perioperative period in diabetic patients.
5.	Elective surgery should be deferred in hyper- or hypothyroid patient till the patient is euthyroid however, mild hypothyroidism should not be a deterrent to postpone the surgery.
6.	Thiopentone is the induction agent of choice in hyperthyroid patients.
7.	At least tachycardia should be settled in hyperthyroid patients undergoing emergency surgery.
8.	As the metabolism of drugs is reduced in hypothyroidism, all drugs should be used in minimal possible doses.
9.	Halothane is absolutely contraindicated in pheochromocytoma because it sensitizes myocardium to catecholamines.
10.	Increase in catecholamines during tumor handing can cause profound hypertension and life-threatening arrhythmias during pheochromocytoma surgery.
11.	Etomidate is contraindicated in adrenal insufficiency because it causes adrenocortical suppression.

Chapter 24 – Anesthesia for Neuromuscular Diseases

1.	For myasthenic patients anticholinesterases should be continued in reduced doses.
2.	Local/regional anesthesia is the anesthetic technique of choice for myasthenia gravis patients.
3.	Desflurane is the most preferred inhalational agent for maintenance in myasthenia gravis patients because it produces maximum muscle relaxation among the currently used inhalational agents.
4.	Myasthenic patients are resistant to depolarizers and hypersensitive to non-depolarizers.
5.	Mivacurium, atracurium, and Cis-atracurium are safe non-deploarizers for myasthenia gravis patients.

6.	Myasthenic syndrome patients are sensitive to both depolarizing and non-depolarizing muscle relaxants.
7.	Succinylcholine is absolutely contraindicated for muscular dystrophies; it may produce cardiac arrest due to hyperkalemia and rhabdomyolysis and can precipitate malignant hyperthermia in these patients.
8.	Non-depolarizers exhibit increased sensitivity in muscular dystrophies making mivacurium, atracurium, cis-atracurium to be safer choices.
9.	Halothane is contraindicated in muscular dystrophies as it can precipitate malignant hyperthermia.

Chapter 25 – Anesthesia for Immune Mediated and Infectious Diseases

1.	Due to atlantoaxial instability intubation in patients with rheumatoid arthritis should be performed with neck in extension.
2.	In patients with ankylosing spondylitis fusion of lumbar spine may make central neuraxial blocks impossible and fusion of cervical spine may make intubation impossible.
3.	CD4 count and viral load do not determine the rate of perioperative anesthetic complications.

Chapter 26 – Anesthesia for Disorders of Blood

1.	Minimum hemoglobin of 8 g/dL is considered optimal to accept the patient for elective surgery.
2.	Considering the complications associated with blood transfusion, the approach in present day practice is towards the minimum use of blood products called as restrictive approach to transfusion. CHAPTER 26: Anesthesia for Disorders of Blood 219
3.	Exchange transfusion is not practiced technique for sickle cell patient in current day. The approach is just to correct the anemia by giving red cell transfusion with the target hematocrit > 30 (Hemoglobin >10 g%).
4.	The most important perioperative goal in sickle cell patients is to avoid hypoxia, acidosis, dehydration, hypothermia, stress and pain.
5.	Intravenous regional anesthesia (Bier’s block) can be utilized for sickle cell patients.
6.	Platelet count of > 50,000/mm3 is recommended before considering the patients for surgery.
7.	The deficit coagulation factor should be increased to at least 50% of the normal before considering the patient for surgery.
8.	Regional anesthesia is contraindicated for patients with deficient coagulation factor disorders like hemophilia.
9.	Administrating methylene blue can be life-threatening in patients with G6PD deficiency.
10.	Barbiturates are absolutely contraindicated for all kind of porphyrias except porphyria cutanea tarda and erythropoietic protoporphyrias.
11.	Propofol, opioids and suxamethonium are considered safe for porphyrias patients.
12.	Fasting, dehydration and infection can trigger acute crisis of porphyria.

Chapter 27 – Neurosurgical Anesthesia

1.	All inhalational agents inhibit autoregulation of cerebral blood flow while it remains largely preserved with most of the intravenous anesthetics.
2.	pCO2 should be maintained between 25–30 mm Hg to achieve optimal hypocapnia.
3.	Hyperosmolar drugs like mannitol are not absolutely contraindicated in disrupted blood brain barrier like head injury or stroke however, should be used judiciously.
4.	All inhalational agents (including nitrous oxide) increase the cerebral blood flow and ICT. The order of vasodilating potency is- halothane >>Desflurane ≈ Isoflurane>Sevoflurane.
5.	Due to more favorable effects sevoflurane is preferred over isoflurane as an inhalational agent of choice in neurosurgery in current day anesthetic practice. CHAPTER 27: Neurosurgical Anesthesia 229
6.	All IV agents except ketamine decrease CBF, CMR and ICT.
7.	As hyperglycemia can cause cerebral edema, Glucose containing solutions should be avoided in neurosurgery.
8.	If necessary colloids can be safely used in neurosurgical patient however, there their use should be limited.
9.	Routine use of hypothermia is not recommended.
10.	Blood sugar should be best maintained between 140–180 mg%.
11.	Capnography is more than mandatory monitor for neurosurgical patients.
12.	The aim during neurosurgical procedures is to maintain cerebral perfusion pressure between 70 and 100 mm Hg.
13.	Not only induction, recovery should also be very smooth in neurosurgical patients.
14.	Steroids must be started in preoperative period in patients having brain edema due to brain tumor while preoperative steroids should be avoided if edema is due to head injury.
15.	Central neuraxial blocks are absolutely contraindicated in raised ICT.
16.	Suxamethonium is not absolutely contraindicated for neurosurgery.
17.	All head injury patients should be assumed to be having cervical injury and basal skull fracture until proved otherwise.
18.	Venous air embolism is the most bothersome complication of posterior fossa craniotomies.
19.	Venous air embolism can become clinically significant if the volume of entertained air > 100 mL.
20.	The most sensitive tool to detect venous air embolism is transesophageal echocardiography.

21.	After stroke surgery should avoided in period of maximal vasospasm risk (4–14 days) as far as possible.
22.	Carotid endarterectomy can be undertaken after 6 weeks while other elective surgeries should be deferred for 9 months after cerebrovascular accident.
23.	The major concerns of spine surgery are related to prone position.

Chapter 28 – Anesthesia for Obstetrics

1.	Hyperdynamic circulation makes pregnant patients more vulnerable for cardiac failure.
2.	Decreased FRC, ERV and increased oxygen requirement makes pregnant patients more vulnerable for hypoxia. CHAPTER 28: Anesthesia for Obstetrics 235
3.	Minimum alveolar concentration decreases by 25 to 40% in pregnant patients making them more susceptible to anesthetic over dosage.
4.	Pregnant patients are very prone to develop high spinal.
5.	Supine hypotension syndrome (SHS) can cause severe hypotension or even cardiac arrest after spinal anesthesia. To prevent SHS the pregnant patient should lie in left lateral position.
6.	All anesthetic drugs except muscle relaxants (only Gallamine has significant transfer) and glycopyrolate can be transferred to fetus from maternal circulation.
7.	Due to alkaline pH local anesthetics and opioids can get accumulated in fetus.
8.	Regional anesthesia is preferred over general anesthesia for cesarean section because risk of pulmonary aspiration and possibility of failed intubation can be avoided.
9.	Because of pulmonary aspiration and failed intubation the mortality in obstetrics during GA is twice as compared to regional anesthesia.
10.	All pregnant patients must be considered full stomach irrespective of their fasting status and must be managed like rapid sequence induction.
11.	At concentration less than half minimum alveolar concentration (MAC) inhalational agents do not produces significant uterine relaxation.
12.	Contrary to old recommendations propofol can be safely used in pregnancy.
13.	Anesthetic technique of choice for PIH is epidural.
14.	Most commonly used technique for painless labor is continuous lumbar epidural.
15.	Maternal demand is sufficient indication to start painless labor.
16.	Low concentration epidurals neither increase the rate of cesarean section nor the rate of instrumental delivery.
17.	Nonobstetric elective surgery should be deferred until postpartum period (6 weeks postdelivery) while urgent surgeries should be performed during the second trimester.

18.	Anesthesia of choice for non-obstetric surgery is regional.

Chapter 29 – Pediatric Anesthesia

1.	The larynx is more anterior and high and subglottis (at the level of cricoid) is the narrowest part in children.
2.	Due to high oxygen requirement and low reserve, children are very prone for developing hypoxia.
3.	Contrary to previous recommendation cuffed tubes can be used in children but should allow a leak at a pressure of more than 15–20 cm H2O.
4.	Bradycardia is the most common adverse cardiac event during perioperative period in pediatric patients.
5.	The high volume of distribution increases the initial dose requirement in children.
6.	MAC requirement in decreasing order is infants > neonates > adults.
7.	A simpler and probably more accurate formula than 4-2-1 is to give 20–40 mL/kg of fluid during intraoperative period. Thereafter, in postoperative period the fluid is given by the formula of 2-1-0.5.
8.	As children are not so prone for hypoglycemia ringer lactate may be the only fluid required.
9.	Inhalational induction is the most commonly utilized induction method in pediatric patients however, the method of choice for induction is always IV.
10.	Due to smooth and rapid induction, sevoflurane (7–8%) is considered as the inhalational agent of choice for induction.
11.	Isoflurane and desflurane has got irritating induction therefore, must not be used for induction.
12.	Sevoflurane and halothane are excellent agents for induction but not suitable for maintenance as they can cause postoperative agitation.
13.	Atracurium and Cisatracurium are the relaxant of choice as they do not depend on hepatic and renal functions.
14.	Succinylcholine should be avoided in children.
15.	Pain perception in pediatric patients is same as adults.
16.	Age is no bar for regional anesthesia; any block can be given at any age.
17.	Bag and mask ventilation is contraindicated for diaphragmatic hernia, tracheoesophageal fistula and pyloric stenosis.
18.	In diaphragmatic hernia positive pressure ventilation should be done with airway pressure < 20 cm H2O.
19.	Nitrous oxide is contraindicated in diaphragmatic hernia surgery.
20.	The position of endotracheal in tracheoesophageal fistula should be below the fistula but above the carina.

21.	Metabolic, fluid and electrolyte correction must be done before taking the patient for pyloric stenosis surgery.
22.	Avoid elective surgeries in premature babies till 50 weeks of their postconception age.
23.	Airway management may be difficult in children suffering from Down syndrome. .

Chapter 30 – Geriatric Anesthesia

1.	Diastolic dysfunction, the most common cardiac finding in old age, is a significant determinant of morbidity.
2.	Chronological age is neither a contraindication for any surgery nor is a criterion to do battery of routine investigations.
3.	Assessment of cognitive function is must in geriatric patients.
4.	Regional anesthesia is preferred over general anesthesia.
5.	The requirement of benzodiazepines and opioids may be reduced by 50%.
6.	The minimum alveolar concentration (MAC) of inhalational agents decreases by 4–6% per decade.
7.	Desflurane because of its rapid emergence and least metabolism is the preferred inhalational agent for geriatric patients.
8.	Atracurium and Cis-atracurium are the muscle relaxant of choice for old age patients.
9.	The overall prevalence of postoperative delirium in older patients is estimated to be 10%.
10.	Short-term changes in cognitive function have been very well-documented in 10–15% of geriatric population after surgery.
11.	The incidence of POCD is independent of both surgery and anesthesia.

Chapter 31 – Anesthesia for Obese Patients (Bariatric)

1.	To perfuse extra fat obese patients are in high cardiac output state.
2.	Obese patients may have extra parenchymal restrictive pattern in their pulmonary functions.
3.	Polysomnography should be strongly considered in morbidly obese patients or in patients with a history of obstructive sleep apnea.
4.	Due to difficult airway management, anesthesia of choice is regional anesthesia.
5.	Airway management is the most challenging part in obese patients.
6.	As the ramped position provides a superior view of laryngoscopy it is preferred position for intubation for obese patients.
7.	Lipid soluble drugs (IV induction agents, opioids) have increased volume of distribution, therefore given on actual body weight basis. Water soluble drugs (nondepolarizing muscle relaxants) have limited volume of distribution, therefore should be given on ideal body weight basis.

8.	Desflurane is the inhalational agent of choice for maintenance in obese patients.
9.	Obese patients are very prone to hypoxia in post- operative period.
10.	Application of CPAP and PEEP improves oxygenation in obese patients.

Chapter 32 – Anesthesia for Laparoscopy

1.	Carbon dioxide is the most frequently used gas for laparoscopy because it is more soluble.
2.	Argon is considered as most ideal gas for laparoscopy. CHAPTER 32: Anesthesia for Laparoscopy 251
3.	Intra-abdominal pressure of 12–14 mm Hg can decrease the cardiac output by 30–35%.
4.	Capnothorax usually resolves by itself in 30–60 minutes; chest tube drainage should be reserved only for the cases not responding to conservative measures or which are life threatening.
5.	CO2 embolism can be fatal complication of laparoscopy; it carries a mortality of 30%.
6.	In CO2 embolism, there is a biphasic response on capnography.
7.	Intra-abdominal pressure of > 20 mm Hg can significantly impair renal, mesenteric and intestinal blood flow. Therefore, intra-abdominal pressure should be kept between 12 and 14 mm Hg.
8.	Nausea and vomiting is the most common postoperative complication of laparoscopic surgeries; incidence can be as high as 40–75%.
9.	General anesthesia with endotracheal intubation and mild hyperventilation is the most commonly used technique for most of the laparoscopic surgeries.
10.	A patient with the controlled ischemic heart disease can be safely considered for laparoscopic surgeries.
11.	Laparoscopy is considered safe during pregnancy.
12.	Laparoscopy once used to be considered as contraindication in obese patients has now become the preferred method over open surgeries.

Chapter 33 – Anesthesia for Ophthalmic Surgery

1.	Topical eye medications can have systemic effects.
2.	One of the most important goals during eye surgery is to reduce or at least maintain the intraocular pressure within normal range.
3.	Traction on extraocular muscle (especially the medial rectus) can precipitate oculocardiac reflex.
4.	Most of the ocular surgeries are performed under regional anesthesia.
5.	Peribulbar block has now almost completely replaced retrobulbar block.
6.	All IV agents (except ketamine) and inhalational agents reduce the IOP.
7.	Nitrous oxide should be avoided in retinal detachment surgery if air, sulfur hexafluoride or perfluoro-propane bubble is used for correction.

8.	Succinylcholine increases the IOP, however the rise is transient (5–10 minutes), therefore if the risk of pulmonary aspiration is high, it is safer to use succinylcholine.

Chapter 34 – Anesthesia for ENT Surgery

1.	In preoperative evaluation of patients posted for panendoscopy, assessment of airway by preoperative endoscopy or indirect laryngoscopy is must.
2.	In the majority (>95%) of the patients it is possible to get the panendoscopy surgery done with MLS tube.
3.	Nitrous oxide should be avoided in laryngeal surgeries because it can increase the post-surgical edema by causing expansion of the cuff of the endotracheal tube.
4.	The most important complication of laser surgery is fires.
5.	Laser resistant endotracheal tubes are not 100% protective against laser fires.
6.	Tonsillar patients should be nursed in tonsillar position in the postoperative period to prevent aspiration and laryngospasm.
7.	Post tonsillectomy bleeding re-exploration patients should be managed like full stomach patient.
8.	Nitrous oxide should not be used in patients with blocked Eustachian tubes and tympanoplasties.

Chapter 35 – Anesthesia for Trauma and Burns

1.	Management of trauma patients poses unique challenges due to difficulty in obtaining a medical history, inadequate fasting, intoxication, multiple injuries and shock.
2.	Systolic blood pressure of at least 80–90 mm Hg is recommended before induction. CHAPTER 35: Anesthesia for Trauma and Burns 261
3.	Invasive blood pressure and temperature monitoring is highly recommended for trauma patients.
4.	Requirement of IV induction doses is drastically reduced in hypotensive patients.
5.	Etomidate and ketamine are the agents often selected for induction.
6.	All trauma patients should be assumed to be having cervical spine injury and basal skull fracture until proved otherwise.
7.	Patients with maxillary fractures can be easily intubated through oral route; patients with mandibular fracture are often intubated nasally. Patients with both mandibular and maxillary fracture may require tracheostomy.
8.	Desflurane is often the most preferred inhalational agent in trauma patients.
9.	Succinylcholine remains the muscle relaxant of choice for intubation if there is risk of aspiration or difficult airway while atracurium and Cis- atracurium are preferred non depolarizers in trauma.
10.	Airway may be difficult in burn patients due to contracture of face and neck.
11.	In burns succinylcholine is contraindicated for one year while dose of nondepolarizers is increased.

Chapter 36 – Anesthesia for Orthopedics

1.	Cardiac arrest can occur at the time of tourniquet deflation.
2.	Pressure in tourniquet should not exceed more than 100 mmHg above systolic pressure and duration should not exceed more than 2 hours for both upper and lower limbs.
3.	The triad of fat embolism syndrome is dyspnea, confusion and petechial hemorrhages. Petechial rash is pathognomonic of FES.
4.	Cement implantation syndrome may be characterized by hypoxia, profound hypotension or even cardiac arrest.
5.	The incidence of fatal PE after hip surgery can be as high as 1–3%.
6.	There have been case reports of stroke or even brain death in shoulder surgeries done in the beach chair position.
7.	Regional anesthesia is always preferred over general anesthesia due to less incidence of thromboembolism, less blood loss, fewer respiratory complications and better control of pain.
8.	The function of the spinal cord is monitored by evoked responses like somatosensory evoked potential (SSEP), motor evoked potential (MEP) and electromyogram (EMG).

Chapter 37 – Anesthesia at Remote Locations

1.	At high altitude, rotameter under reads the actual flow rate.
2.	Vaporizer settings need not be changed at low or high barometric pressure, except for desflurane.
3.	The incidence of postspinal headache is high at high altitude.
4.	Nitrous oxide should not be used in helium atmosphere.
5.	Small and crowded space, nonavailability of adequate equipment, monitors, technical support and help in case of emergency, inadequate preoperative evaluation of patients and inadequate knowledge of anesthesia intricacies by medical proceduralists make nonoperating room anesthesia very challenging and least preferred by majority of anesthetists.
6.	Full fasting recommendations are required for non-operating room anesthesia.
7.	Capnography is highly desirable for nonoperating room anesthesia.
8.	Most often, either sedation with midazolam or total intravenous anesthesia (TIVA) with Propofol + Remifentanil is utilized for nonoperating room procedures.
9.	For MRI suites, anesthesia machine, monitors and oxygen cylinders should be made up of aluminum.

Chapter 38 – Anesthesia for Day-care Surgery

1.	Duration of surgery, the American Society of Anesthesiologist (ASA) grading and age (except prematures) are not considered as absoulte parameters to deny a surgery on day-care basis.
2.	Midazolam is the benzodiazepine of choice for day-care patients.
3.	Nil orally guidelines for day-care patients are same as for indoor patients.

4.	GA with laryngeal mask airway (LMA) is always preferred over intubation.
5.	For day-care patients, propofol is the most preferred agent for induction, desflurane for maintenance and mivacurium for muscle relaxation.
6.	The combination of choice for TIVA is Propofol + Remifentanil.
7.	Spinal anesthesia is frequently used for day-care patients, however, with small gauge pencil tip needles and short/intermediate acting agents.
8.	The aim of discharge should be fast-track discharge.
9.	The two most common causes of unexpected admission are uncontrolled pain and nausea and vomiting.
10.	Able to accept liquids and pass urine is not considered as mandatory criteria for discharging a patient after day care surgery.
11.	Most often the re-admission is due to surgical complications such as surgical site infection.

Chapter 39 – Pain Management

1.	Visual analog scale is the most commonly used scale for pain assessment.
2.	Postoperative pain management should have multimodal approach.
3.	Opioid remains the mainstay of treatment for post-operative analgesia.
4.	Epidural route is the most preferred route for postoperative analgesia.
5.	Patient-controlled analgesia (PCA) is the method of choice if opioids have to be given by parenteral route.
6.	Regional nerve blocks with local anesthetics as a single shot or continuous infusion provide excellent postoperative analgesia.
7.	Radiofrequency ablation (RFA) is always preferred over chemical neurolysis.
8.	Transforamina route for the delivery of steroids is preferred over epidural route.
9.	Radiofrequency (RF) ablation of medial branch of posterior ramus of spinal nerve is the most definitive modality of treatment for facet arthropathy.
10.	Most common neuropathy encountered in clinical practice is diabetic neuropathy.
11.	Drug of choice for neuropathic pain is pregabalin followed by gabapentin.
12.	Trigeminal neuralgia still responds best to carbamazepine.
13.	For upper limb CRPS, stellate ganglion block is performed; and, for lower limb dystrophies, lumbar sympathetic chain block is performed.
14.	Opioid remains the mainstay of treatment for cancer pain.
15.	The multiple trigger points are pathognomonic of fibromyalgia.

16.	Highest blood level of local anesthetic is achieved after intercostal nerve block.

Chapter 40 – Intensive Care Management

1.	Oxygen therapy is helpful in improving oxygen saturation in all types of hypoxia except histotoxic hypoxia.
2.	Ventilatory breath may be volume controlled or pressure controlled.
3.	The chief advantage of volume controlled ventilation is decreased chances of hypoventilation and chief disadvantage is increased chances of barotrauma.
4.	The chief advantage of pressure controlled ventilation is less chances of barotrauma while main disadvantage is increased possibility of hypoventilation.
5.	Pressure regulated volume control (PRVC) can be considered as mode of choice for ventilation in current day critical care practice.
6.	Ideal PEEP is the PEEP which maintain oxygen saturation >90% without decreasing the cardiac output significantly. Usually, it is between 5–12 cm H_2O.
7.	Leakage is the major problem during NIPPV.
8.	It is possible to wean patient in any mode of ventilation except control mode ventilation.
9.	The most common source of nosocomial infections in ICU are urinary tract infections.
10.	Majority of nosocomial infections in ICU are caused by gram-negative organisms.
11.	Positing head at 30 degrees proves to be simplest and most cost effective method to decrease ventilator associated pneumonia.
12.	Current guidelines for hemorrhagic shock are in favor of hypotensive resuscitation which means maintaining SBP between 80–100 mm Hg and MAP between 60–65 mm Hg.
13.	Correction of coagulopathy called as hemostatic resuscitation is of utmost importance in hemorrhagic shock.
14.	The concept of hypotensive and hemostatic resuscitation is called as balanced resuscitation.
15.	Modified shock index (MSI), ratio of heart rate to mean blood pressure (MAP) is considered as better predictor of outcome in shock then shock index.
16.	Mixed venous oxygen saturation is overall the best guide to assess tissue perfusion while urine output is considered as best clinical guide to assess tissue perfusion.
17.	Stroke volume variation is considered as most reliable to assess fluid status and titrate fluid therapy.
18.	Crystalloids are preferred over colloids for shock and Ringer is the most preferred crystalloid.
19.	Phenylephrine is the vasopressor of choice for neurogenic shock, ephedrine for spinal shock and nor-epinephrine for septicemic shock.
20.	Metabolic acidosis due to shock should be treated by fluids and vasopressors not by soda bicarbonate unless severe (pH <7.15).

21.	Enteral route is always preferred over parenteral route for nutrition.
22.	Subclavian is vein of choice for parenteral nutrition.
23.	Out of the total energy required 60–70% should be provided by carbohydrates (dextrose) and 30–40% by fats. Amino acids are given as supplement, never as a substrate for source of energy.
24.	Sepsis is the leading cause of death in ICU.
25.	New studies do not recommended tight glycemic control in septicemia.
26.	Low pO_2, decreased functional residual capacity (FRC) and decreased static compliance of lungs are the hallmark of ARDS while pCO_2 may be low, normal or high.
27.	Low tidal volume (6 mL/kg), airway plateau pressure < 30 cm H_2O and low FiO_2 (<0.5) are the key in the management of ARDS.
28.	Extracorporeal membrane oxygenation (ECMO) should be instituted early for refractory ARDS.
29.	In current day practice the use of recruitment measures, prone position, inhaled nitric oxide and muscle relaxants are reserved for refractory ARDS.
30.	For COPD patients it is necessary to keep the low flows (1–2 L/min) of oxygen to preserve hypoxic drive.
31.	Maintaining oxygenation by non-invasive positive pressure ventilation (NIPPV) with CPAP/BIPAP should be the prime goal in COPD patients.
32.	Ventilator setting for COPD patients includes small tidal volume, low breath rate (6–8/min.) and longer expiratory time to allow maximum exhalation.
33.	Lactic acidosis is the most common cause of metabolic acidosis in anesthesia and is due to tissue hypoxia.
34.	Soda bicarbonate should be used to treat only severe acidosis (pH <7.15).

Chapter 41 – Cardiopulmonary and Cerebral Resuscitation

1.	Unwitnessed cardiac arrest in adults should be considered due to ventricular fibrillation until proved otherwise and in children should be considered due to asystole until proved otherwise.
2.	The standard sequence of CPR, i.e. A (airway) → B (breathing) → C (circulation) was changed to C → A → B in 2010 CPR guidelines.
3.	Epiglottis, rather than tongue fall, has been found to be the more important cause of airway obstruction in unconscious patient.
4.	In patients with suspected cervical spine fracture if airway is not manageable by jaw thrust alone then head tilt and chin lift can be given.
5.	For patients with cervical spine fractures intubation should be performed with neck in neutral position and only after stabilization. Manual stabilization is preferred over collar stabilization.
6.	Rescue breath by mouth to mouth is reserved only for trained rescuers.

7.	For untrained lay rescuer CPR means “hands only” i.e. only cardiac massage.
8.	Intubation is the most definitive and best method for ventilation.
9.	Cardiac massage should be able to depress sternum by 2 inches (5 cm) but not more than 2.4 inches (6 cm).
10.	During CPR rescuer must not lean on patient chest between compressions.
11.	The rate of cardiac compression should be 100–120 compressions/minute but not more than 120. CHAPTER 41: Cardiopulmonary and Cerebral Resuscitation 311
12.	Without advanced airway the ratio of compression to breathing for adults should be 30:2 irrespective of number of resuscitators while for children this ratio is 30: 2 for single rescuer and 15:2 for two rescuers.
13.	With advanced airway compression is continued at a rate of 100–120 compressions/minute and breathing at a rate of 10 breaths/minute without any synchronization.
14.	In the current day CPR Capnography is considered essential.
15.	Successful cardiac massage should be able to produce $ETCO_2$ of at least 20 mm Hg.
16.	Ventricular fibrillation, pulseless ventricular tachycardia and polymorphic ventricular tachycardia are shockable rhythms while pulseless electrical activity and asystole are considered as nonshockable rhythms.
17.	Shock energy should be 360J with monophasic defibrillators and 150–200 with biphasic defibrillators.
18.	Immediately after giving shock give 5 cycles of 30:2 or 2 minutes of CPR before checking the rhythm.
19.	The end point to stop CPR is the return of spontaneous circulation (ROSC) assessed by palpating carotids.
20.	Vasopressin, which was given after 1st dose of adrenaline, has been removed from 2015 CPR guidelines.
21.	Defibrillation should be attempted 2 minutes after epinephrine and Amiodarone injection.
22.	Consider termination of CPR efforts if no response for 20 minutes.
23.	Hypovolemia (shock) is the most common cause of pulseless electrical activity (PEA).
24.	Targeted temperature management is the only recommended method for brain protection.
25.	Targeted temperature management means maintaining core body temperature between 32–36° for 24 hours.
26.	Prognostication is recommended after 72 hours.
27.	In unwitnessed cardiac arrest in children 5 cycles of CPR are given before activating EMS.

28.	During CPR in pregnancy aortocaval compression should be prevented by uterine displacement, not by lateral tilt.
29.	Emergency cesarean section should be considered if fetus >25 week and cesarean section can be performed within 4 minutes of maternal arrest.
30.	Intraosseous route is recommended for all ages, anything can be given and is preferred over endo-tracheal route.
31.	Fluid of choice during CPR is normal saline.
32.	Adrenaline by intra-cardiac route is contraindicated in present day CPR.
33.	The routine empiric use of sodium bicarbonate is not recommended.
34.	The intravenous/intraosseous/endotracheal tube (IV/IO/ET) concentration of adrenaline during CPR is 1: 10,000.
35.	Amiodarone is considered as drug of choice for ventricular tachycardia.